

Team Update 02

Date: 2026-09-26 · **Amends:** Version 2.2 as released at kickoff (TU-00) and Team Update 01

The Game Manual plus every Team Update is the current ruleset; where they conflict, the most recent update governs. Each change below gives the rule or section, the old text struck through, the new text in bold, and a one-line rationale. New sections and tables are given in full as added text. The published documents (the Markdown and PDF manual, the brief, the Field CAD Package, the drawings, the vision guide and the Onshape generator) already incorporate this update.

Summary. The FIELD gains **SCREE**: 18 SCREE PATCHES of low half-round ridges, nine in each half, laid in a staggered pattern around each ALLIANCE's staging marks, which ROBOTS either drive over or weave between (new Section 3.8 and Table 3-7). The **BASE DEPOT** becomes a closed ring around all four faces of each CRAG: four legs and four corner squares, 81.5 in square over the lip, with 4096 in² of floor, room for about 18 CACHE CRATES or 27 mixed SUPPLIES (was 8 / 12). The ring holds every ROBOT's FRAME PERIMETER 19.75 in off every CRAG face, so a ROBOT may now extend up to **30 in** beyond its FRAME PERIMETER while any part of its BUMPERS intersects its own ALLIANCE's CRAG APRON, and 18 in everywhere else (**G404, R105**). The binding reach is a High Peg tip, 26.68 in away. The FIELD centerline and the CENTER CACHE band's side lines are broken for the ring. DEPOT point values (AUTO 4 / TELEOP 2), the single-layer SCORED test, the Regional CAMP I substitution, the SUPPLY LINE RP thresholds (15 / 19 / 23), the rule text of **G502** and every violation are unchanged. No rule is added, removed or renumbered. The Onshape generator now builds SCREE and the ring, so a field built with an earlier generator must be rebuilt, and the manual's figures are re-rendered from the rebuilt field model.

A. Game Manual

A1. Section 2.2, The FIELD (new sentence at the end of the paragraph)

The open floor around each ALLIANCE's staging marks is broken up by **SCREE**, patches of low ridged terrain that ROBOTS either drive over or weave between.

Rationale: introduces SCREE (A14).

A2. Section 2.2, BASE DEPOT

A ROBOT may possess at most 2 SUPPLIES at a time. Each CRAG ~~also has a floor-level BASE DEPOT tray~~ is also **ringed at floor level by a BASE DEPOT** tray that accepts any SUPPLY in any orientation.

Rationale: the BASE DEPOT now surrounds the CRAG (A11).

A3. Section 2.4, Table 2-3, BASE DEPOT row

any SUPPLY, any orientation (~~~12~~) (**~27**)

Rationale: the ring's single-layer capacity (A11). The DEPOT values are unchanged.

A4. Section 3, opening paragraph

The SUMMIT PUSH ARENA includes all elements of the game infrastructure required to play a MATCH: the FIELD, two CRAGS, two HEADWALLS, four OUTFITTERS, **18 SCREE PATCHES**, 63 SUPPLIES, and the equipment for scorekeeping, field control, and MATCH lighting.

Rationale: SCREE is part of the ARENA.

A5. Section 3.1 (new sentence at the end of the first paragraph)

In each half of the FIELD, nine SCREE PATCHES of low ridged terrain lie on the carpet around the alliance staging marks (Section 3.8).

Rationale: as A4.

A6. Section 3.2, Table 3-1, FIELD centerline and CENTER CACHE band rows

FIELD centerline: 2-in line at $X = 324$ running the full width, $Y 0-324$, interrupted where ~~the two CRAG footprints cross it (the BASE DEPOT trays stop 8 in short of the line)~~ **each CRAG and its BASE DEPOT cross it, between the inner edges of that CRAG's APRON tape: Blue $Y 198-282$, Red $Y 42-126$**

CENTER CACHE band: $X 300-348$, $Y 108-216$; **its ends coincide with the two SOCKET FACE planes, under the BASE DEPOTS, so no end lines are laid, and its two side lines stop at the inner edges of the APRON tape, like the centerline ($Y 126-198$)**

Rationale: the ring crosses $X = 324$ and covers both ends of the CENTER CACHE band. Breaking both lines at the inner edge of the APRON tape leaves no stub of tape between the tape and the DEPOT's 1.0-in entry chamfer, whose outer edge is at Blue $Y 198.25$ and 281.75 .

A7. Section 3.2, CRAG APRON paragraphs

The CRAG APRON is a line-call placement protection at each ALLIANCE's own CRAG: a ROBOT whose BUMPERS intersect its own ALLIANCE's APRON may not be contacted by an opponent ~~(G407): (G407), and may extend up to 30 in beyond its FRAME PERIMETER instead of 18 in, over that APRON (G404).~~

~~The APRON offset is 36 in on the SHELF FACE and the PEG FACE and 20 in on the two SOCKET FACES. At a SOCKET FACE, the shallower offset limits protection. A socket rim stands 8.0 in out from its face, so a ROBOT using the full 18 in extension has its FRAME PERIMETER 26 in out and its BUMPERS 23 in out: 3 in beyond the 20-in tape, and unprotected. To be protected at a SOCKET FACE, a ROBOT must engage the rim with 15 in of extension or less. At the SHELF FACE and PEG FACE, with their 36-in offset, every reachable position is protected.~~

The APRON offset is 36 in on the SHELF FACE and the PEG FACE and 20 in on the two SOCKET FACES. At a SOCKET FACE, the shallower offset limits how far out a ROBOT can stand and stay protected. The BASE DEPOT lip holds a ROBOT's BUMPERS 16.75 in off every CRAG face, inside the APRON, and from there each socket rim, 8.0 in out from its face, is 11.75 in of extension away. A ROBOT that stands farther off stays protected, and keeps the 30-in extension of G404, only while its BUMPERS still intersect the 20-in tape: its FRAME PERIMETER is then no more than 23 in out, and the rim no more than 15 in of extension away. A ROBOT reaching the rim from farther out is outside the APRON, unprotected, and held to 18 in; at the full 18 in its BUMPERS are 23 in out, 3 in beyond the tape. At the SHELF FACE and PEG FACE, with their 36-in offset, every position from which a ROBOT can reach a scoring feature is protected.

Rationale: the APRON now also scopes the extension limit (A21), and the DEPOT lip, not the face, sets the standoff at every face.

A8. Section 3.3, opening paragraph and Table 3-2

Each CRAG has a 48 × 48 in footprint and rises to a spire top 90 in above the ~~carpet~~; **with the BASE DEPOT tray that rings its base (Section 3.3.4), it occupies an 81.5-in square over the lip (83.5 in over the entry chamfer).**

Table 3-2, SHELF FACE features: ~~Shelf 1, Shelf 2, the Summit Socket, the BASE DEPOT Shelf 1, Shelf 2, the Summit Socket~~

New sentence after Table 3-2: **The BASE DEPOT runs along the base of all four faces.**

Rationale: the BASE DEPOT is no longer a SHELF FACE feature.

A9. Section 3.3.2, socket standoff

~~The two sockets on a face have different approach standoffs. The BASE DEPOT's corner arm runs 16.0 in along each SOCKET FACE from the SHELF FACE, and the Low Socket sits directly above it, so a ROBOT servicing the Low Socket parks against the arm and reaches 11.75 in. The Mid Socket, on the PEG FACE side, is clear of the arm and is 5.0 in of reach from BUMPERS on the face. Both are inside the R105 limit.~~

Both sockets on a SOCKET FACE stand over the BASE DEPOT's leg along that face (Section 3.3.4): each rim center is 8.0 in out, inside the 16.0-in channel, and 14.0 in lateral, inside the 48-in leg. A ROBOT servicing either socket therefore parks with its BUMPERS against the DEPOT lip, its FRAME PERIMETER 19.75 in off the face, and reaches 11.75 in to the rim, the same reach as the Summit Socket needs from the SHELF FACE. All five socket rims are well inside the G404 extension limit.

Rationale: the SOCKET FACE leg now runs under the Mid Socket too, so its reach rises from 5.0 to 11.75 in.

A10. Section 3.3.3 (new paragraph after the ROPE COIL rest pose)

The BASE DEPOT's PEG FACE leg runs beneath the Low and Mid Pegs, so a ROBOT servicing the PEG FACE parks with its BUMPERS against the DEPOT lip and its FRAME PERIMETER 19.75 in off the face. A peg projects 7.07 in horizontally from its face (10.0 in at 45°), so a Low or Mid Peg tip is 12.68 in of extension away and its root 19.75 in. The High Pegs stand on the spire face, 14.0 in inboard of the PEG FACE, so a High Peg tip is 26.68 in away and its root 33.75 in. The tip is within the 30-in extension a ROBOT has while its BUMPERS are on its own CRAG APRON (G404), and it is the longest reach the CRAG requires. The root is beyond reach, so a ROPE COIL is placed on a High Peg by releasing it over the tip and letting it drop onto the peg, where it settles at the root as described above. Releasing a SUPPLY from rest is not LAUNCHING (G502).

Rationale: the PEG FACE leg moves the PEG FACE standoff from 3.0 to 19.75 in, which sets the new extension limit.

A11. Section 3.3.4, BASE DEPOT

The BASE DEPOT is a floor tray running along the base of the SHELF FACE and wrapping 16.0 in around both of that face's corners onto the SOCKET FACES. Its floor sits 0.25 in above the carpet, and its lip top is 4.0 in above the carpet. The channel is 16.0 in deep, measured from the CRAG face. In plan it is a continuous open-topped U: a 16 × 80 in outer run parallel to the SHELF FACE (the 16 × 48 in shelf face leg plus a 16 × 16 in corner square at each end), with a 16 × 16 in corner arm running back along each SOCKET FACE.

The BASE DEPOT is a floor tray that surrounds the CRAG on all four faces. Its floor sits 0.25 in above the carpet, and its lip top is 4.0 in above the carpet. The channel is 16.0 in deep, measured from each CRAG face to the inner face of the lip. The lip is 0.75 in thick with a 0.25-in radius on its top edge, and a 1.0-in 45° entry chamfer runs around its outside. In plan the tray is one continuous open-topped ring of four legs, each 16 × 48 in, one along each face (the SHELF FACE leg, the two SOCKET FACE legs, and the PEG FACE leg), joined by a 16 × 16 in corner square at each of the four corners.

New second paragraph: At the outer faces of the lip the ring is an 81.5-in square centered on the CRAG (Blue X 283.25–364.75, Y 199.25–280.75; Red X 283.25–364.75, Y 43.25–124.75), and the chamfer brings it to 83.5 in. It lies inside the CRAG APRON on every face. On the SOCKET FACES, where the APRON is only 20 in deep, the chamfer's outer edge comes within 0.25 in of the APRON tape along the face and within 0.16 in at the corner arcs. The ring crosses the FIELD centerline, which is broken for it (Section 3.2).

The shelves overhang 14.0 in of the channel and span only the 48 in width of the SHELF FACE. From above, the outer 2.0 in of the shelf face leg is open, as are both outer corner squares and part of each corner arm. Each arm is overhung by its Low Socket tube, so only the corner squares are wide enough to drop a CACHE CRATE or a ROPE COIL straight in. The rest of the shelf face leg is loaded by pushing SUPPLIES in over the lip;

The shelves overhang 14.0 in of the SHELF FACE leg and span only the 48-in width of the SHELF FACE. From above, the outer 2.0 in of the SHELF FACE leg is open, as are all four corner squares, the PEG FACE leg, which only the 1.5-in Low and Mid Pegs cross, and both SOCKET FACE legs except beneath their socket tubes. A SUPPLY cannot be dropped into the tray where a socket tube overhangs it. The rest of the SHELF FACE leg is loaded by pushing SUPPLIES in over the lip; Shelf 1 and its gussets stay above Z = 19.0, clear of a crowned CACHE CRATE standing on the tray floor (top at Z = 13.25).

Capacity: ... capacity is a single-layer packing limit: ~~roughly 8 CACHE CRATES~~; **the tray has 4096 in² of floor, room for about 18 CACHE CRATES one abreast**, or about ~~12~~ **27** SUPPLIES in a mixed load.

Rationale: the ring ($80^2 - 48^2 = 4096$ in² of floor, up from 1792) replaces the U. Crates cannot sit two abreast in the 16-in channel, so each 80-in run holds six and each 48-in leg between the runs three. The SCORED test, the DEPOT values and the SUPPLY LINE thresholds are unchanged.

A12. Section 3.3.6, CRAG APRON

G402, G403, G404, G501, and G502 also refer to it.

New paragraph: The APRON also sets the extension limit. A ROBOT may extend up to 30 in beyond its FRAME PERIMETER while any part of its BUMPERS intersects its own ALLIANCE's CRAG APRON, provided every part more than 18 in out stays over that APRON (inside its outer tape edge, which takes in the BASE DEPOT and the CRAG); everywhere else the limit is 18 in (G404, R105). The BASE DEPOT holds the FRAME PERIMETER of every ROBOT servicing the CRAG 19.75 in off the face, and the longest reach from there is 26.68 in, to a High Peg tip (Section 3.3.3). The larger allowance stops at the APRON tape because the HEADWALL climb depends on the 18-in limit: with 30 in, a ROBOT on the carpet could wrap the CAMP RUNG (18.02 in) and the SUMMIT RUNG (24.45 in) without hanging from a lower rung (G416). An opponent's APRON gives a ROBOT neither protection nor extra extension. The extra reach is for crossing the DEPOT, not for reaching out of the APRON into the corridor or the CENTER CACHE.

Rationale: states the scoped limit where the APRON is defined (A21).

A13. Section 3.7, Mounting heights

Each CRAG face carries a pair of tags centered ± 14 in from the face ~~centerline~~: **centerline, with the BASE DEPOT tray beneath them on every face.**

Rationale: the 0.19-in CACHE CRATE margin and the upright-CELL case now apply on all four faces. The tag poses and the layout JSON are unchanged.

A14. New Section 3.8, SCREE, and Table 3-7

3.8 SCREE

SCREE is rough terrain laid on the carpet in the open floor around each ALLIANCE's staging marks. It is made up of 18 SCREE PATCHES, nine in each half of the FIELD, set out in a staggered, non-linear pattern. A ROBOT crossing that floor either drives over SCREE, which is slow and a tip risk for a tall ROBOT with a raised or extended MECHANISM, or slows down and weaves between the patches.

New Figure 3-6. SCREE in the Blue half of the FIELD, seen from above the Blue alliance wall: the nine SCREE PATCHES around the alliance staging marks, with the two CRAGS and their BASE DEPOT rings beyond. Dimensioned drawing: Plate 1 of the field drawing set.

- Patch: each SCREE PATCH is a 30.0 × 30.0 in square with its sides parallel to the FIELD axes, sitting directly on the carpet with no base plate.
- Ridges: each patch carries four half-round ridges of 1.75-in radius, so each ridge is 1.75 in tall and 3.5 in wide at its base. The ridges run straight across the patch and are cut square at its edges. Their centerlines lie 4.0 in and 12.0 in either side of the patch center, measured square to the ridges: an 8.0-in pitch, with 4.5 in of flat carpet between adjacent ridges.
- Ridge direction: "+45" ridges run parallel to the line $X = Y$, "-45" ridges parallel to the direction (1, -1), and "Y" ridges parallel to the Y axis, square to the main direction of travel down the FIELD. A Red patch has the same ridge direction as its Blue twin, because a 180° rotation preserves the direction of a line.
- Material: matte HDPE half-round rod fastened to the carpet, in a dark warm grey distinct from the carpet and from both ALLIANCE colors (03-field/MATERIALS-AND-COLORS.md).

Table 3-7: SCREE PATCH centers and ridge directions

Patch	Blue center (X, Y)	Red center (X, Y)	Ridges
S1	(105, 88)	(543, 236)	+45
S2	(105, 162)	(543, 162)	Y
S3	(105, 236)	(543, 88)	-45
S4	(200, 37)	(448, 287)	+45
S5	(200, 125)	(448, 199)	+45
S6	(200, 199)	(448, 125)	-45
S7	(200, 287)	(448, 37)	-45
S8	(234, 108)	(414, 216)	+45
S9	(234, 216)	(414, 108)	-45

Each Red patch is its Blue twin rotated 180° about the FIELD center. Each half is also mirror-symmetric about $Y = 162$, so both OUTFITTERS of an ALLIANCE face the same SCREE. In the Blue half the patches stand in a near column at X 90–120 (S1–S3), a far column at X 185–215 (S4–S7), and an inner pair at X 219–249 (S8, S9). The alliance staging marks at $X = 144$ lie between the near and far columns. Every patch is dimensioned on Plate 1 of the field drawing set and in the FIELD CAD PACKAGE §1.4.

Gaps. Neighboring patches leave gaps that a ROBOT can weave through: 44.0 in between S1 and S2, S2 and S3, and S5 and S6; 58.0 in between S4 and S5, and S6 and S7; 41.2 in corner to corner between S4 and S8, and S7 and S9; and 65.4–68.3 in between the near and far columns. Two kinds of gap are closed. S5 and S8, and S6 and S9, stand only 4.0 in apart and read as two L-shaped clusters, and S4 and S7 leave only 22.0 in to the guardrail.

Flat ground. In the Blue half, no SCREE lies in the strip X 48–90 across Y 90–234 in front of the HEADWALL, which leaves 42 in of flat carpet for squaring up to climb; in a 36 × 24 in pad at each alliance staging mark; in the

mouth of each OUTFITTER LANE (X 48–90); or in the strip X 252–264 up to the CRAG APRON line at X = 264. The Red half mirrors these at X 558–600 and X 384–396. SCREE lies outside every protected zone (the CRAG APRONS, the BASECAMP / HEADWALL ZONES, and the OUTFITTER LANES) and outside the CENTER CACHE, and it occludes no AprilTag, since every tag is mounted at 12 in or higher.

SCREE is part of the FIELD. ROBOTS may drive over it, and G202 applies to it as to every FIELD element. A SUPPLY that comes to rest on SCREE stays in play and is not restaged. SCREE does not change the BUMPER ZONE (R403).

Commentary: Crossing SCREE takes at least 2.0 in of clearance under the whole frame and wheels of about 6 in or larger. A conventional swerve drive, with 3–4-in wheels and about 1.25–1.5 in of frame clearance, high-centers on a ridge, and so does a stock kit chassis; low-clearance ROBOTS slow down and weave through the gaps instead. BUMPERS at the top of the R403 range, with the bottom edge at 2.5 in, clear a ridge by 0.75 in on flat carpet and by about 0.5 in with a wheel riding a ridge; BUMPERS set lower in that range meet the ridges. The layout leaves few straight lines. Of the straight 28-in-wide paths from the OUTFITTER LANE mouths and the BASECAMP front to the SHELF FACE and SOCKET FACE approaches, only one narrow corridor that a ROBOT can drive misses every patch: in the Blue half, it leaves the BASECAMP front with its centerline at Y ≈ 175 (its upper edge at Y ≈ 189) and runs to the far (+Y) corner of the SHELF FACE approach, with about 0.4 in to spare. Every other cycle route weaves around SCREE or crosses it. A ROBOT rocks as each wheel climbs a ridge, so a tall ROBOT that crosses at speed with an elevator raised or an arm extended can tip; lowering the MECHANISM first, or going around, avoids the risk.

Rationale: organizer decision: rough terrain around the staging marks makes drivetrain choice and routing part of the design problem. SCREE is kept clear of every protected zone, the CENTER CACHE, the HEADWALL approach and the OUTFITTER LANE mouths.

A15. Section 4.3.5, AUTO Distances (new sentence at the end of the first paragraph)

SCREE (Section 3.8) lies across the routes from BASECAMP and from the staging marks, so those distances understate the distance a ROBOT actually drives.

Rationale: the published distances stay straight-line distances.

A16. Section 4.4.4 (new paragraph)

SUPPLIES on SCREE: SCREE is part of the FIELD (Section 3.8). A SUPPLY that comes to rest on it stays in play, and FIELD STAFF do not move or restage it.

Rationale: a SUPPLY on SCREE is neither out of bounds nor restored.

A17. Section 4.4.5, Cycle Distances and Table 4-7

New sentence at the end of the first paragraph: **Each approach position is a reference point on the outer edge of the BASE DEPOT channel, 16.0 in off the middle of its face; a ROBOT servicing the face stops with its BUMPERS at the lip, 16.75 in off the face, and its center about half its BUMPER length farther out.**

Table 4-7: OUTFITTER (24, 294) → PEG FACE approach (~~368, 240~~) **(364, 240)** | ~~348 in~~ **344 in**

New paragraph after Table 4-7: **These are straight-line distances, and they understate real cycles: SCREE (Section 3.8) lies across most of the routes from the OUTFITTERS and the staging marks, so a ROBOT either detours around the patches or slows down to cross them.**

The CRAG has 17 scoring positions in total (6 shelf slots, 5 sockets, 6 pegs) plus a BASE DEPOT that holds ~~roughly 12~~ **about 27** SUPPLIES.

Rationale: every approach position is now defined the same way on every face; the PEG FACE point moves 4 in to match the SHELF FACE and SOCKET FACE points.

A18. Section 4.6, Scoring Summary

A CRAG filled to capacity in TELEOP is worth 107 placement points (28 low + 49 mid + 30 high); a full BASE DEPOT adds about ~~24~~: **54**.

Rationale: 27 SUPPLIES at the TELEOP DEPOT value of 2.

A19. Section 4.10, ALLIANCE Roles, RING ALPINIST row

Hook or spear end effector plus a 78-in ~~reach~~: **reach that extends about 27 in beyond the FRAME PERIMETER (§3.3.3).**

Rationale: the High Peg tip is now the binding horizontal reach (A10).

A20. G202 *Do not damage the ARENA*, Commentary (new sentence after the first)

SCREE is a FIELD element built to be driven over (Section 3.8), so driving across its ridges does not violate this rule either; prying up or tearing loose a ridge does.

Rationale: G202 applies to SCREE as to every FIELD element.

A21. G404 *Extend no more than 18 in*

~~**G404 *Extend no more than 18 in.***~~ A ROBOT may not extend more than 18 in horizontally beyond its FRAME PERIMETER at any point during a MATCH (see **R105**).

G404 *Extend no more than 18 in, or 30 in on your own CRAG APRON.* A ROBOT may not extend more than 18 in horizontally beyond its FRAME PERIMETER at any point during a MATCH, except that while any part of its BUMPERS intersects its own ALLIANCE's CRAG APRON it may extend up to 30 in, provided every part of it more than 18 in beyond the FRAME PERIMETER is over that APRON: inside the outer edge of the APRON tape, which takes in the BASE DEPOT and the CRAG itself (see **R105**). Extension is measured from the vertical projection of the FRAME PERIMETER. A SUPPLY in a ROBOT's CONTROL (**G501**) is not counted when measuring extension.

The violation is unchanged. New Example and Commentary:

Example: A Blue ROBOT with its BUMPERS against the DEPOT lip reaches 26.7 in to release a ROPE COIL over a High Peg tip. Its BUMPERS are inside the Blue CRAG APRON, so the reach is legal. It then backs away without retracting, and its BUMPERS clear the APRON tape while its arm is still 26 in out: MINOR FOUL. A Red ROBOT defending on the Blue APRON is not on its own APRON and is held to 18 in throughout.

Commentary: The BASE DEPOT holds the FRAME PERIMETER of every ROBOT servicing the CRAG 19.75 in off the face, and from there a High Peg tip is 26.68 in away (Section 3.3.3), beyond an 18-in limit. The 30-in allowance is scoped to the APRON because the HEADWALL climb depends on the 18-in limit: with 30 in anywhere, a ROBOT on the carpet could wrap the CAMP RUNG (18.02 in) and the SUMMIT RUNG (24.45 in) without first hanging from a lower rung (**G416**). The HEADWALL is far from every APRON, so the 18-in limit governs there. The allowance is for reaching across the BASE DEPOT, not out into the FIELD: extension beyond 18 in must stay over the ROBOT's own APRON, so a ROBOT backed onto a SOCKET FACE APRON cannot reach 30 in into the corridor or the CENTER CACHE, which stay open to defense (**G407**).

Rationale: with the ring under every face, a High Peg tip is 26.68 in of extension away, beyond 18 in. A blanket 30-in limit would break the HEADWALL climb ladder: the CAMP RUNG (18.02 in to wrap) and the SUMMIT RUNG (24.45 in) would come within reach from the carpet, so the larger allowance applies only on the ROBOT's own CRAG APRON, which never reaches a HEADWALL.

A22. G416, first Commentary

... wrapping the CAMP RUNG needs 18.02 in, just over the 18-in limit; **limit that applies everywhere outside a ROBOT's own CRAG APRON**, and the SUMMIT RUNG needs 24.5 in, both measured past the far face of the rung.

Rationale: the HEADWALL reach analysis still rests on 18 in.

A23. G502 *No LAUNCHING*, Commentary

The rule text is unchanged. The commentary is replaced:

~~The exception is drawn from two lines the FIELD already carries: the APRON tape and the centerline. Both CRAGS straddle $X = 324$, so the half of a ROBOT's own APRON on its SHELF FACE side of the centerline is $X < 324$ for Blue and $X > 324$ for Red. That half contains the SHELF FACE and the whole tray. Outside it, any LAUNCHED SUPPLY is a violation, and referees need no trajectory judgment. Inside it, flicking a SUPPLY over the 4-in DEPOT lip is legal and intended. Shelf, socket, and peg scoring must be by placement, because testing placement is the purpose of those scoring positions. The exception stops at the centerline because, from the PEG FACE, the tray is on the far side of a 90-in spire.~~

The exception is drawn from two lines the FIELD already carries: the APRON tape and the centerline. Both CRAGS straddle $X = 324$, so the half of a ROBOT's own APRON on its SHELF FACE side of the centerline is $X < 324$ for Blue and $X > 324$ for Red. Outside that half, any LAUNCHED SUPPLY is a violation, and referees need no trajectory judgment. Inside it, flicking a SUPPLY over the 4-in DEPOT lip is legal and intended. Shelf, socket, and peg scoring must be by placement, because testing placement is the purpose of those scoring positions.

The BASE DEPOT rings the CRAG and crosses the centerline, so the exception half does not contain the whole tray. It contains the SHELF FACE, the SHELF FACE leg and its two corner squares, and it runs along both SOCKET FACES up to the centerline. From those SOCKET FACE parts of the half, the SOCKET FACE legs and the two PEG FACE corner squares lie in a direct line along the face, and a SUPPLY LAUNCHED from there that comes to rest anywhere in the tray is within the exception. Only the middle of the PEG FACE leg is out of line, behind the CRAG's 60-in body. The exception still stops at the centerline, the line referees already watch: beyond it, the APRON lies in the opponent's half of the FIELD, and the SHELF FACE half already reaches every part of the tray but the middle of the PEG FACE leg, which a ROBOT on the PEG FACE side fills by pushing SUPPLIES in or releasing them from rest.

Rationale: the ring crosses the centerline, so the old statement that the exception half contains the whole tray no longer holds, and the tray is no longer behind the spire from the PEG FACE.

A24. R105 *Extend no more than 18 in*

~~**R105** *Extend no more than 18 in.* ROBOTS may not extend more than 18 in (45.7 cm) horizontally beyond their FRAME PERIMETER at any time during a MATCH, in any direction, measured as a horizontal distance from the vertical projection of the FRAME PERIMETER.~~

R105 *Extend no more than 18 in, or 30 in on your own CRAG APRON.* ROBOTS may not extend more than 18 in (45.7 cm) horizontally beyond their FRAME PERIMETER at any time during a MATCH, in any direction, measured as a horizontal distance from the vertical projection of the FRAME PERIMETER, except that while any part of its BUMPERS intersects its own ALLIANCE's CRAG APRON a ROBOT may extend up to 30 in (76.2 cm), provided every part of it more than 18 in beyond the FRAME PERIMETER is over that APRON (inside the outer edge of the APRON tape). A ROBOT that can extend more than 18 in must be built or programmed to hold 18 in whenever its BUMPERS are outside its own CRAG APRON, and to keep any extension beyond 18 in over that APRON.

Violation: ROBOT will not pass INSPECTION; maximum extension is **extension, and the means of holding 18 in outside the CRAG APRON and of keeping longer reach over it, are** demonstrated at INSPECTION per Section 7.2. Over-extension during a MATCH is penalized under **G404**.

Commentary: The FIELD is dimensioned so that 18 in of extension reaches every scoring position on the CRAG. A ROBOT with its BUMPERS against the DEPOT lip has its FRAME PERIMETER 19.75 in from the SHELF FACE plane, so a Shelf 1 slot center is 12.75 in of extension away, and the Summit Socket rim, 8.0 in outboard of that face at 72 in, is 11.75 in away. From the PEG FACE, a High Peg root is 17.0 in away and its tip only 9.9 in. The two sockets on a SOCKET FACE are reached from different standoffs. The Mid Socket sits on the PEG FACE side, clear of the DEPOT, so BUMPERS can come to the face and the rim is 5.0 in of extension away. The Low Socket sits on the SHELF FACE side above the DEPOT's corner arm, which holds the FRAME PERIMETER 19.75 in off the face and puts that rim 11.75 in away. Nothing on the CRAG requires more than 18 in; the design problem is vertical reach.

Commentary: The BASE DEPOT rings the CRAG, so a ROBOT servicing any face parks with its BUMPERS against the DEPOT lip, inside its own CRAG APRON, and its FRAME PERIMETER 19.75 in off the face. From there a Shelf 1 slot center is 12.75 in of extension away; the Summit Socket rim and every Low and Mid Socket rim, each 8.0 in out from its face, are 11.75 in away; a Low or Mid Peg tip is 12.68 in away and its root 19.75 in; and a High Peg tip, on the spire face 14.0 in inboard of the PEG FACE, is 26.68 in away, with its root at 33.75 in. The High Peg tip is the binding reach, at 26.68 of the 30 in allowed on the APRON, and a ROPE COIL is released over it to drop onto the peg (Section 3.3.3). The 30-in allowance stops at the APRON tape because a blanket 30-in limit would let a ROBOT wrap the CAMP RUNG (18.02 in) and the SUMMIT RUNG (24.45 in) from the carpet and bypass the HEADWALL climb (G416). The HEADWALL is far from every APRON, so the 18-in limit governs its reach analysis. Every scoring position except a High Peg is within 19.75 in of the FRAME PERIMETER, so the main design problem is vertical reach.

Rationale: as A21. Reach from BUMPERS at the lip, all horizontal: shelf slot 12.75, every socket rim 11.75 (the Mid Socket was 5.0), Low and Mid Peg tips 12.68 (roots 19.75), High Peg tip 26.68 (root 33.75).

A25. R106, Commentary

Teams accept the corresponding risk: a fully extended, high-CG ROBOT may be defended wherever defense is legal (see G4xx), **SCREE rocks a ROBOT that drives over it (Section 3.8)**, and tall ROBOTS tip.

Rationale: SCREE adds a tip risk for tall ROBOTS.

A26. Section 7.2, Sizing

Inspectors also ask the team to demonstrate the ROBOT's maximum horizontal extension and verify that it does not exceed ~~18~~ **30** in beyond the FRAME PERIMETER (R105). **A ROBOT that can extend more than 18 in must also show how it holds 18 in whenever its BUMPERS are outside its own CRAG APRON: either a software limit tied to the ROBOT's position on the FIELD, or a MECHANISM that can exceed 18 in only toward the CRAG.**

Rationale: INSPECTION checks the new maximum and how the ROBOT holds 18 in off its APRON.

A27. Section 7.4, Table 7-1, items 4 and 9

Item 4, Extension limit: CAD views of each MECHANISM at its commanded maximum reach, with a dimension showing ~~≤18~~ **≤30** in of horizontal extension beyond the FRAME PERIMETER. **For any reach over 18 in, a statement of how the ROBOT holds 18 in outside its own CRAG APRON and keeps any reach beyond 18 in over that APRON (a software limit tied to field position, or a MECHANISM that exceeds 18 in only toward the CRAG).**

Item 9, Reach proof: Measure from the standoff the FIELD imposes: ~~19.75 in from the SHELF FACE; from a SOCKET FACE, 19.75 in at the Low Socket (the ROBOT cannot drive over the BASE DEPOT corner arm below it) and 3.0 in at the Mid Socket; and 3.0 in from the PEG FACE.~~ **the BASE DEPOT holds the FRAME PERIMETER 19.75 in off every CRAG face, so a High Peg tip is 26.68 in of extension away and its root is beyond reach (Section 3.3.3).**

Rationale: as A21 and A24.

A28. Section 9, Glossary

APRON (CRAG APRON): ... unless that ROBOT initiated the contact (~~G407~~): **(G407)**, and such a ROBOT may extend up to 30 in beyond its FRAME PERIMETER instead of 18 in, over that APRON (G404).

BASE DEPOT (short form: **DEPOT**): The floor tray with a 4-in lip and a 16-in-deep channel that ~~runs along the SHELF FACE and wraps 16 in around both of that face's corners onto the SOCKET FACES.~~ **surrounds a CRAG on all four faces: a ring of four legs, one along each face, joined by four corner squares, 81.5 in square over the lip.** It accepts any SUPPLY in any orientation in a single layer (~~roughly 8 CACHE CRATES, or about 12 SUPPLIES mixed~~), **(about 18 CACHE CRATES, or about 27 SUPPLIES mixed)**, and a SUPPLY counts only when the tray floor alone supports it, wholly inside the channel's vertical projection.

SHELF FACE: The CRAG face that faces the owning ALLIANCE's wall, carrying ~~Shelf 1, Shelf 2, the Summit Socket, and the BASE DEPOT.~~ **Shelf 1, Shelf 2, and the Summit Socket.**

New entries:

SCREE	The rough terrain on the carpet around each ALLIANCE's staging marks: 18 SCREE PATCHES, nine in each half of the FIELD, in a staggered pattern (Section 3.8). SCREE is part of the FIELD; ROBOTS may drive over it, and a SUPPLY that comes to rest on it stays in play.
SCREE PATCH	One 30.0 × 30.0 in patch of SCREE, carrying four straight half-round ridges 1.75 in tall at an 8.0-in pitch, set at 45° to the FIELD axes or parallel to the Y axis (Table 3-7).

Rationale: as A7, A11 and A14.

A29. Figures 2-1 and 3-1 to 3-5 (editorial), and new Figure 3-6

Figures 2-1 and 3-1 to 3-5 are re-rendered from the rebuilt field model, which shows SCREE, the BASE DEPOT ring and the broken FIELD centerline and CENTER CACHE band side lines. The figures of Team Update 01 showed the U-shaped DEPOT. Figure 3-6 (Section 3.8, A14) is new.

Rationale: the figures now show the field as modeled.

A30. Appendix A plates (Drawings 1 to 4 and 6)

- **Drawing 1 (FIELD top view):** all 18 SCREE PATCHES with their ridges, a SCREE panel with the patch table and a ridge section, and a SCREE key symbol; the BASE DEPOT drawn as a ring; the FIELD centerline and the CENTER CACHE band side lines broken at the inner edges of the APRON tape (Blue Y 198–282, Red Y 42–126); no CENTER CACHE band end lines; the plan notes follow.
- **Drawing 2 (CRAG):** every view shows the ring, with the four legs and four corner squares in View D (81.5 over the lip) and the reach notes of A10 and A24 (High Peg tip 26.68 of the 30 in allowed). In D4, the lip thickness reads **0.75, +0 / -0.25** (B5).
- **Drawing 3 (HEADWALL):** the climb-reach table and its note state that the 18-in limit governs at the HEADWALL, because the 30-in allowance applies only on a ROBOT's own CRAG APRON.
- **Drawing 4 (OUTFITTER), editorial:** the crate-through-chute envelope dimension reads "13.0 crowned", so the label no longer runs past its extension lines.
- **Drawing 6 (AprilTag map):** the occlusion note states that the DEPOT runs below every CRAG face, and that an upright O2 CELL covers the bottom 0.81 in of the target.

Rationale: the plates carry A6–A14 and A21–A24.

B. Field CAD Package, materials, vision guide and Onshape generator

B1. Field CAD Package §1, element table

BLUE CRAG and RED CRAG: 48 × 48 in footprint, spire top 90 in in; with its BASE DEPOT ring, an 81.5-in square, X 283.25–364.75, Y 199.25–280.75 (Red: Y 43.25–124.75)

New row: SCREE | 18 SCREE PATCHES, 9 per half; centers in §1.4 | 30.0 × 30.0 in each, four ridges 1.75 in tall

B2. Field CAD Package §1.2, tape

CRAG APRON (new sentence at the end of the row): The tape is laid inside the offset line, so its inner edge is 18.0 in off each SOCKET FACE; the BASE DEPOT ring lies inside it with only 0.25 in to spare along the SOCKET FACES and 0.16 in at the corner arcs (§3, APRON clearance check).

FIELD centerline: ... broken where the two CRAG footprints cross it (the BASE DEPOT trays stop 8 in short of the line); each CRAG and its BASE DEPOT ring cross it. The break runs between the inner edges of that CRAG's APRON tape, Blue Y 198–282, Red Y 42–126, so that no stub of line is left between the APRON tape and the DEPOT's entry chamfer (plan envelope Blue Y 198.25–281.75, Red Y 42.25–125.75). G402 and G502 are both line calls against it.

CENTER CACHE band: Its end edges coincide with the two CRAG SOCKET FACE planes, and it is interrupted where the BASE DEPOT trays cross it, which lie under the BASE DEPOT rings, so no end lines are laid. Its two side lines (X 300–302 and X 346–348) stop, like the FIELD centerline, at the inner edges of the two APRON tapes, so they run Y 126–198.

Rationale (B1–B2): as A6 and A11.

B3. Field CAD Package, new §1.4 SCREE

SCREE is rough terrain on top of the carpet, laid in the open floor around each alliance's staging marks: 18 SCREE PATCHES, nine per half, in a staggered, non-linear pattern. A ROBOT either drives over a patch, which is slow and a tip risk for a tall ROBOT with a raised or extended mechanism, or slows down and weaves between patches. SCREE is part of the FIELD: ROBOTS may drive over it, G202 applies to it, and a SUPPLY that comes to rest on it stays in play and is not restaged (the Game Manual governs).

Feature	Dimension	Status
Patch footprint	30.0 × 30.0 in square, sides parallel to the field axes, sitting directly on the carpet (no base plate)	CRITICAL
Ridges per patch	4, straight, running across the whole patch and cut square (vertically) at the patch boundary	CRITICAL
Ridge profile	half-round, radius 1.75 in: 1.75 in tall and 3.5 in wide at the base, flat side on the carpet	CRITICAL
Ridge centerlines	±4.0 and ±12.0 in from the patch center, measured along the ridge normal: an 8.0-in pitch, with 4.5 in of flat carpet between adjacent ridges	CRITICAL
Ridge direction	per patch (table below): +45 = parallel to direction (1, 1), the line $X = Y$; -45 = parallel to (1, -1); $Y =$ parallel to the Y axis, across the main +X direction of travel	CRITICAL
Material	HDPE half-round rod fastened to the carpet, matte, color scree (MATERIALS-AND-COLORS.md)	(ref)

SCREE PATCH table (centers CRITICAL). Red twin = (648 - x, 324 - y). A 180° rotation preserves line direction, so a Red patch has the same ridge direction as its Blue twin.

Patch	Blue center	Blue footprint	Ridges	Red center (twin)
S1	(105, 88)	X 90–120, Y 73–103	+45	(543, 236)
S2	(105, 162)	X 90–120, Y 147–177	Y	(543, 162)
S3	(105, 236)	X 90–120, Y 221–251	-45	(543, 88)
S4	(200, 37)	X 185–215, Y 22–52	+45	(448, 287)
S5	(200, 125)	X 185–215, Y 110–140	+45	(448, 199)
S6	(200, 199)	X 185–215, Y 184–214	-45	(448, 125)
S7	(200, 287)	X 185–215, Y 272–302	-45	(448, 37)
S8	(234, 108)	X 219–249, Y 93–123	+45	(414, 216)
S9	(234, 216)	X 219–249, Y 201–231	-45	(414, 108)

Each half is also mirror-symmetric about $Y = 162$: S1/S3, S4/S7, S5/S6 and S8/S9 are mirror pairs with opposite ridge directions, and S2 lies on the axis. Both OUTFITTERS of an alliance therefore face identical SCREE.

Layout checks (verify in CAD). These properties were checked numerically against the table; a model that disagrees with them is misplaced. Coordinates are Blue; Red is the 180° rotation.

- Columns: near column X 90–120 (S1–S3), far column X 185–215 (S4–S7), inner pair X 219–249 (S8, S9). The staging marks at $X = 144$ lie between the near and far columns.
- Passable gaps, through which a ROBOT can weave: 44.0 in (S1–S2, S2–S3, S5–S6), 58.0 in (S4–S5, S6–S7), 41.2 in corner to corner (S4–S8, S7–S9), and 65.4–68.3 in between the near and far columns.
- Closed gaps: 4.0 in (S5–S8, S6–S9, which read as two L-shaped clusters) and 22.0 in between S4 / S7 and the guardrail.

- Flat ground kept clear of SCREE: X 48–90 across Y 90–234 in front of the HEADWALL (42 in for squaring up to climb); a 36 × 24 in pad at each staging mark; each OUTFITTER LANE mouth (X 48–90); and X ≥ 252 up to the CRAG APRON line at X = 264. Red is mirrored: X 558–600 in front of its HEADWALL, and X ≤ 396 back to its APRON line at X = 384.
- Non-linear: one narrow corridor gives a clear straight path 28 in wide from the BASECAMP front to a CRAG approach: path centerline Y ≈ 175 at the BASECAMP front (X = 48; upper edge Y ≈ 189), running to the far (+Y) corner of the SHELF FACE approach at X = 264 (centerline Y ≈ 265–269), with about 0.4 in to spare over the 14 in of clearance each side of the centerline. Every other drivable straight 28-in path from an OUTFITTER LANE mouth or the BASECAMP front to the SHELF FACE or either SOCKET FACE approach crosses a patch, so every other cycle route must weave or cross SCREE.
- Zones and tags: SCREE lies outside every protected zone (the CRAG APRONS, the BASECAMP / HEADWALL ZONES and the OUTFITTER LANES) and outside the CENTER CACHE. It occludes no AprilTag: its ridges stand 1.75 in tall, and the lowest tag panel edge on the FIELD is at Z = 7.50 (the HEADWALL lane panels, §7).

Robot interaction (reference). A BUMPER with its bottom edge at 2.5 in, the top of the range R403 allows, clears a ridge by 0.75 in on flat carpet and by about 0.5 in with a wheel riding a ridge; R403 also allows a bottom edge as low as 1.25 in, which meets the ridges. Crossing SCREE needs at least 2.0 in of clearance under the whole frame and wheels of about 6 in or larger. Conventional swerve (3–4-in wheels, about 1.25–1.5 in of frame clearance) high-centers on it, so low-clearance ROBOTS, including a stock kit chassis, weave between the patches instead.

CAD notes: model each ridge as a half-round extrusion (R1.75 profile, flat side on the carpet) long enough to cross the patch at its offset and direction, then clip it to its 30 × 30 in patch square with a vertical cut. Model S1, S2, S4, S5 and S8, mirror S1, S4, S5 and S8 about Y = 162 for S3, S7, S6 and S9 (the mirror flips +45 to –45), then rotate-pattern the Blue half 180° about (324, 162) for Red. Robot-driving dims: patch 30.0, ridge height 1.75 (R1.75), pitch 8.0, offsets ±4.0 / ±12.0, patch centers and ridge directions per the table.

Rationale: the modeling specification for A14.

B4. Field CAD Package §2, CRAG

- **§2.1, new bullet: BASE DEPOT ring: the tray (§3) surrounds the tower on all four faces, so in plan each CRAG with its DEPOT occupies an 81.5-in square over the lip (83.5 in over the entry chamfer), centered on the CRAG.**
- **§2.2: The SHELF FACE also carries the Summit Socket (§2.4) and the BASE DEPOT (§3). (§2.4). The BASE DEPOT's SHELF FACE leg (§3) runs along its base, as a DEPOT leg does along every face.**
- **§2.3, Robot standoff: Robot standoff: the two sockets on a face are reached from different distances.** The BASE DEPOT's corner arm runs 16.0 in along each SOCKET FACE from the SHELF FACE plane, and the Low Socket, at -14.0 lateral, sits directly above it. A ROBOT servicing the Low Socket therefore parks against the arm's outer lip: the lip outer face is 16.75 in off the face, the BUMPER face is there, and the FRAME PERIMETER is 3.0 in behind it at **19.75 in**, so the rim (8.0 in outboard of the face) is **11.75 in of horizontal extension** away. The Mid Socket, at +14.0 lateral, is clear of the arm: BUMPERS come to the face, the FRAME PERIMETER is 3.0 in off it, and the rim is **5.0 in** away. Both are inside the 18-in limit, but a mechanism designed only for the 5.0 in case cannot reach the Low Socket. Model the arm and the Low Socket together and check this before committing to an arm length. **Robot standoff: both sockets on a face are reached across the BASE DEPOT. The DEPOT's SOCKET FACE leg (§3) runs the full 48-in length of each SOCKET FACE, and both sockets stand directly above it. A ROBOT servicing either socket therefore parks against the leg's outer lip: the lip outer face is 16.75 in off the face, the BUMPER face is there, and the FRAME PERIMETER is 3.0 in behind it at 19.75 in, so each rim (8.0 in outboard of the face) is 11.75 in of horizontal extension away, the same for the Low Socket and the Mid Socket (reach table in §3). Both tubes overhang the leg, and a SUPPLY cannot be dropped into the tray where a tube overhangs it (§3, open-to-sky check). Model the SOCKET FACE leg and both sockets together.**
- **§2.4, Robot reach: ... so the rim is 11.75 in of horizontal extension away, inside the 18-in limit: limit (reach table in §3).**
- **§2.5, Robot approach: With BUMPERS against the tower's PEG FACE, the FRAME PERIMETER sits 3.0 in off it, so the Low and Mid Peg tips reach **4.07 in inside** the FRAME PERIMETER's vertical projection at Z = 30 and Z = 54. That is legal, because the pegs are not a ROBOT extension, but a ROBOT approaching this face must either carry an open pocket above its BUMPERS at those heights or stand off until its FRAME PERIMETER clears the tips, at which point a peg root is 7.07 in of reach away. The High Pegs are the opposite case. Their roots sit on the spire's peg face, **14.0 in inboard** of the tower's, so from BUMPERS on the tower face a High Peg root is **17.0 in** away and its tip **9.93 in**. The High Peg root is the tightest horizontal reach on the CRAG, at 17.0 of the 18.0 in allowed. The BASE DEPOT's PEG FACE leg (§3) runs along this face, so BUMPERS stop at its lip, 16.75 in off the face, and the Low and Mid Peg tips stop 9.68 in short of the lip's outer face. From the FRAME PERIMETER, 3.0 in behind the BUMPER face at 19.75 in, a Low or Mid Peg tip is 12.68 in of horizontal extension away and its root 19.75 in. The High Pegs' roots sit on the spire's peg face, 14.0 in inboard of the tower's, so a High Peg tip is 26.68 in away and its root 33.75 in. The binding reach on the CRAG is the High Peg tip, at 26.68 of the 30 in allowed while any part of the ROBOT's BUMPERS intersects its own ALLIANCE's CRAG APRON (G404, R105), which it always does with its BUMPERS at the lip. The High Peg root is beyond reach, so a ROPE COIL is placed on a High Peg by releasing it over the tip and letting it drop onto the peg, where it settles in the rest pose above; release from rest is not LAUNCHING (G502).**
- **§2.7, Feature order: ... → depot tray ring (§3) → ...**

Rationale: as A9, A10 and A21.

B5. Field CAD Package §3, BASE DEPOT

Heading: BASE DEPOT (~~tray, ×1 per CRAG~~) (ring tray, ×1 per CRAG)

A floor-level tray wrapping the SHELF FACE and its two adjacent corners: an open-topped U in plan, 16.0 in wide throughout. Its three named parts are used consistently across this package: the **shelf-face leg**, 16 × 48 in, directly outboard of the SHELF FACE (Blue: X 284–300, Y 216–264); two **corner squares**, 16 × 16 in, continuing that leg past each SHELF FACE corner (Blue: X 284–300, Y 200–216 and Y 264–280); together with the leg they form one straight 16 × 80 in outer run; two **corner arms**, 16 × 16 in, turning back along each SOCKET FACE (Blue: X 300–316, same two Y bands). The five rectangles form one continuous channel. Without the corner squares, the leg and the arms would meet only at points, which would be neither a U nor buildable as one weldment.

A floor-level tray surrounding the CRAG on all four faces: a closed square ring in plan, 16.0 in wide throughout. Its named parts are used consistently across this package: four legs, 16 × 48 in, one directly outboard of each face: the SHELF FACE leg (Blue: X 284–300, Y 216–264), the two SOCKET FACE legs (Blue: X 300–348, Y 200–216 and Y 264–280) and the PEG FACE leg (Blue: X 348–364, Y 216–264); four corner squares, 16 × 16 in, one at each CRAG corner (Blue: X 284–300 and X 348–364, each at Y 200–216 and Y 264–280). The eight rectangles form one continuous channel, Blue X 284–364 × Y 200–280 less the CRAG footprint. There are no corner arms. Two opposite legs and the four corner squares form two straight 16 × 80 in runs, joined by the other two legs.

Table rows:

Feature	Old	New
Channel depth	16.0 in	16.0 in, off every face (CRITICAL)
Corner squares and arms Ring layout	channel continues 16.0 in along each SOCKET FACE, measured from the SHELF FACE plane, at the same 16.0 in depth out from that face, plus the two 16 × 16 in outer corner squares that close the U (Blue: X 284–300 × Y 200–216 and X 284–300 × Y 264–280)	four 16 × 48 in legs, one along each face (SHELF FACE, both SOCKET FACES, PEG FACE), and four 16 × 16 in corner squares, one at each corner, closing the ring (Blue channel: X 284–364 × Y 200–280, less the CRAG footprint X 300–348 × Y 216–264) (CRITICAL)
Lip thickness	0.75 in (ref)	0.75 in, no plus tolerance (+0 / -0.25); held for APRON tape clearance, not free to float like a (ref) value
Outer footprint (new row)		81.5 in square over the lip's outer faces (48 + 2 × 16.75), centered on the CRAG: Blue X 283.25–364.75, Y 199.25–280.75; Red X 283.25–364.75, Y 43.25–124.75. The entry chamfer strip brings the plan envelope to an 83.5-in square: Blue X 282.25–365.75, Y 198.25–281.75; Red X 282.25–365.75, Y 42.25–125.75. (ref): follows from the CRITICAL channel and the lip (+0 / -0.25), so it may shrink by up to 0.5 in but never grow
Tray floor	... A 45° entry chamfer strip with a 1.0-in leg runs outside the lip.	lip all the way around, mitred at the four corners.
Capacity	about 8 CACHE CRATES one abreast in the 16-in channel, or about 12 SUPPLIES in a mixed load, in 1792 in² of tray. Twelve crowned crates would need 2028 in² and do not fit.	about 18 CACHE CRATES one abreast in the 16-in channel, or about 27 SUPPLIES in a mixed load, in $80^2 - 48^2 = 4096$ in ² of tray. Crates cannot sit two abreast ($2 \times 13.0 > 16.0$), so the crate count is set by length: each 80-in run holds six crates and each 48-in leg between the runs three.

New check: **APRON clearance check (verify in CAD)**. The ring lies inside every CRAG APRON, but with little room along the **SOCKET FACES**. The chamfer strip's outer edge is 17.75 in off each face. The 2-in APRON tape is laid

inside its 20-in offset line, so the tape's inner edge is 18.0 in off each SOCKET FACE. That leaves 0.25 in along the SOCKET FACES and 0.16 in at the four corner arcs, where the chamfer corner lies 17.84 in from the arc center (Blue (284, 216) at the SHELF FACE / -Y SOCKET FACE corner) against the tape's R18 inner arc. A lip built 0.25 in over its nominal 0.75 in would bring the chamfer onto the tape along the SOCKET FACES and about 0.11 in over it at the corner arcs, so the lip thickness carries no plus tolerance (+0 / -0.25): model it at its nominal 0.75 in. Off the SHELF FACE and the PEG FACE the tape's inner edge is 34.0 in out, far from the ring.

Open-to-sky check. The shelves cantilever 14.0 in from the SHELF FACE and span only that face's 48-in width, so three regions of the tray are open from above. They differ in what they accept: the outer **2.0 in** of the shelf face leg: a 48 × 2 in strip, too narrow for any SUPPLY; both **16 × 16 in outer corner squares** (Blue: X 284–300 × Y 200–216 and Y 264–280): **fully open to the sky**, and therefore the only places where any SUPPLY can be dropped straight in; both **16 × 16 in corner arms** along the SOCKET FACES: open except where the Low Socket tube overhangs them. On the Blue +Y face the tube's plan silhouette covers X 306.7–313.3 × Y 265.6–274.9 from Z 22.19 upward, almost centered in the arm, leaving clear vertical columns of only 6.66, 2.66, 5.11 and 1.56 in. A CACHE CRATE or a ROPE COIL cannot be dropped into an arm from above; it must be pushed in from the adjacent corner square, which is why the square has to be part of the channel. The rest of the shelf face leg is loaded by pushing SUPPLIES in over the lip and under the shelves ... The **G502** launch exception is exercised over the corner squares and the open outer strip.

Open-to-sky check. The shelves cantilever 14.0 in from the SHELF FACE and span only that face's 48-in width, the socket tubes overhang both SOCKET FACE legs, and only the thin Low and Mid Pegs cross the PEG FACE leg. The parts of the ring differ in what they accept from above:

- the SHELF FACE leg: only its outer 2.0 in is open, a 48 × 2 in strip too narrow for any SUPPLY. The rest is loaded by pushing SUPPLIES in over the lip and under the shelves: the Shelf 1 underside is at Z = 23.25, and no gusset descends below Z = 19.0 (the gusset floors in §2.2), against a crowned CRATE's top at Z = 13.25 on the tray floor.
- all four 16 × 16 in corner squares: fully open to the sky.
- both SOCKET FACE legs: open except where the two socket tubes overhang them. On the Blue +Y face the Low Socket tube's plan silhouette covers X 306.7–313.3 × Y 265.6–274.9 from Z 22.19 upward, and the Mid Socket tube's covers X 334.7–341.3 over the same Y band from Z 46.19 upward. Beside each tube the clear columns are only 1.56 in wide toward the face and 5.11 in toward the lip, and a SUPPLY cannot be dropped into the tray where a tube overhangs it. The leg is open between the two tubes (a 21.32-in column, Blue X 313.34–334.66) and beyond each tube, where a 6.66-in column joins the adjacent corner square into a 22.66 × 16 in opening.
- the PEG FACE leg: open except under the four 1.5-in Low and Mid Pegs (Blue Y 226 and Y 254), which reach 7.07 in out over its inner edge. The clear column between the two pegs of a tier is 26.5 in wide.

A crate still fits inside the channel's vertical projection everywhere. The G502 launch exception (BUMPERS in the SHELF FACE half of the own CRAG APRON, Blue X < 324) covers a SUPPLY that comes to rest anywhere in the tray. That half of the APRON contains the SHELF FACE, the SHELF FACE leg and its two corner squares, and it runs along both SOCKET FACES up to the centerline. From the SOCKET FACE parts of it, the SOCKET FACE legs and the PEG FACE corner squares (Blue X 348–364.75, Y 199.25–216 and Y 264–280.75) are in direct line along the face. Only the middle of the PEG FACE leg (Blue X 348–364.75, Y 216–264) lies behind the CRAG, and the 60-in tower body blocks it.

Robot standoff: the lip outer face is 16.75 in from the CRAG face; with the bumper face there, the FRAME PERIMETER is 3.0 in behind it at **19.75 in**. Shelf slot centers are 7.0 in outboard of the face (mid-depth of the 14.0-in cantilevered shelf), so a shelf placement is **12.75 in** of extension; the Summit Socket rim, 8.0 in outboard of the face, is **11.75 in**. Both are inside the 18-in limit.

Robot standoff and reach (reference; all horizontal). The lip's outer face is 16.75 in off every CRAG face. With the BUMPER face there, the FRAME PERIMETER is 3.0 in behind it at 19.75 in, and every scoring position on the CRAG is reached across the tray:

Scoring position	Out from its face	Extension from the FRAME PERIMETER
Shelf slot center (Shelf 1 and Shelf 2; mid-depth of the 14.0-in shelf)	7.0	12.75
Summit Socket rim center	8.0	11.75
Low and Mid Socket rim centers	8.0	11.75
Low and Mid Peg tips / roots	7.07 / 0	12.68 / 19.75
High Peg tips / roots (on the spire's peg face, 14.0 in inboard)	-6.93 / -14.0	26.68 / 33.75

A ROBOT with its BUMPERS at its own CRAG's lip always has them inside its own CRAG APRON, so the 30-in extension limit of G404 and R105 applies there, for reach over that APRON (every CRAG scoring position is inside it); the 18-in limit applies everywhere else on the FIELD. The binding reach is the High Peg tip, at 26.68 of the 30 in allowed. The High Peg root is beyond reach, so a ROPE COIL is placed on a High Peg by releasing it over the tip and letting it drop onto the peg (§2.5). Every other position is within 20 in, and all but the Low and Mid Peg roots are within 18 in.

CAD notes: model the tray as its own sub-assembly mated to the CRAG base, because entrants will study lip-crossing geometry in isolation. ~~Pattern the corner square and corner arm by mirroring one end. Robot-driving dims: lip height 4.0, lip radius 0.25, channel width 16.0, corner square 16.0, corner arm 16.0.~~ The floor, the lip and the chamfer strip are each a square ring (an outer square less an inner square) centered on the CRAG's vertical axis. The tray is symmetric under a 90° rotation about that axis, so one leg and one corner square can be patterned four times. Robot-driving dims: lip height 4.0, lip radius 0.25, channel width 16.0 off every face, corner square 16.0.

Rationale: as A11, A21 and A23. The lip's plus tolerance is removed because the chamfer clears the SOCKET FACE APRON tape by only 0.25 in (0.16 in at the corner arcs).

B6. Field CAD Package §4.1, Climb reach (new sentence at the end of the paragraph)

The HEADWALL is far from every CRAG APRON, so the 18-in limit always applies here; the 30-in allowance of G404 and R105 applies only while a ROBOT's BUMPERS intersect its own CRAG APRON.

Rationale: the rung reach table, and the "Within R105?" column, still use 18 in.

B7. Field CAD Package §7, occlusion budget

- ... **0.19 in** below the target. ~~This is the tightest margin in the package,~~ **The tray runs beneath the tag panels on all four faces, so this margin applies to every CRAG tag. It is the tightest occlusion margin in the package,** so the CRAG tag panel center height is held to **±0.15 in** at field setup ...
- ... an O2 CELL stood on its end in the ~~DEPOT~~ **DEPOT, in front of any of the four faces** (its top at 14.25 in, tray floor 0.25 plus 14.0, against a 13.44-in target ~~bottom~~); **bottom, so it covers the bottom 0.81 in of the target**; ... New sentence at the end of the paragraph: **SCREE (§1.4) occludes no tag: its ridges stand 1.75 in tall, and the lowest panel edge on the FIELD is a HEADWALL lane panel's, at Z = 7.50.**

Rationale: as A13.

B8. Field CAD Package §10, master dimension ledger

#	Element	Old	New
9	BASE DEPOT	lip height / lip radius / channel width / corner square / corner arm / tray floor top: 4.0 · 0.25 · 16.0 · 16.0 · 16.0 · Z = 0.25	lip height / lip radius / channel width / layout / tray floor top: 4.0 · 0.25 · 16.0 off every face · closed ring around all four faces: four 16 × 48 legs + four 16 × 16 corner squares (81.5 square over the lip, (ref)) · Z = 0.25
17 (new)	SCREE PATCH		footprint / ridges / ridge height / pitch / ridge centerlines: 30.0 × 30.0, sides parallel to the field axes · 4, half-round R1.75, cut square at the boundary · 1.75 · 8.0 · ±4.0, ±12.0 from the patch center
18 (new)	SCREE placement		Blue centers (ridge direction) · Red: S1 (105, 88) +45 · S2 (105, 162) Y · S3 (105, 236) -45 · S4 (200, 37) +45 · S5 (200, 125) +45 · S6 (200, 199) -45 · S7 (200, 287) -45 · S8 (234, 108) +45 · S9 (234, 216) -45 · Red = (648 - x, 324 - y), same ridge direction

B9. Field CAD Package §11, drawing list, item 1

field-top-view.svg: ... staging marks, apron offsets, axes, ~~and coordinate callouts~~. **coordinate callouts, and a SCREE panel with the patch table and a ridge section.**

B10. MATERIALS-AND-COLORS.md

- **§1.2 tokens:** depot now covers the BASE DEPOT ~~lip and tray lip~~, **tray floor and entry chamfer (the ring around each CRAG)**. New token: scree | #5C5650 | **SCREE ridges: a dark warm grey, distinct from the carpet and from both alliance colors.**
- **§2 element table:** new row **SCREE ridge | §1.4 | HDPE half-round rod, fastened to the carpet | scree | matte | R1.75 half-round: 3.5 in wide × 1.75 in tall**. BASE DEPOT lip: Painted ~~plywood~~ **plywood, a closed square ring around the CRAG**. DEPOT entry chamfer: Painted plywood ~~strip~~ **strip, mitred at the four corners.**
- **Render checklist:** Both CRAGS with all 17 scoring positions modeled and the BASE DEPOT ~~trays~~ **ring around all four faces of each**; new item **All 18 SCREE PATCHES, each with its four ridges in the direction the FIELD-CAD-PACKAGE.md §1.4 table gives**; the FIELD ~~centerline~~, **centerline broken at each CRAG**, the CENTER CACHE band **side lines** and marks.

B11. VISION-GUIDE.md

- §1.3, first paragraph (new sentence): The BASE DEPOT tray runs beneath the tag panels on all four faces of each CRAG, so every DEPOT row in the table below applies to all eight tags on a CRAG.
- §1.3, new paragraph after the table: SCREE (FIELD-CAD-PACKAGE §1.4) occludes no tag. Its ridges stand 1.75 in tall, and the lowest tag panel edge on the FIELD is at Z = 7.50 (the HEADWALL lane panels), so every sight line from a camera above the ridges passes over them.
- §1.3, 12-in paragraph: At a 12-in center height, the SHELF FACE pair and the alliance-wall-side tag on each SOCKET FACE **every CRAG tag** would be blocked by any CACHE CRATE standing in the BASE DEPOT, **DEPOT beneath it**, which is a routine game state.
- §1.3, upright-CELL case: This affects ~~four~~ **all eight** of each CRAG's ~~eight tags (the two on the SHELF FACE and the alliance-wall-side tag on each SOCKET FACE);~~ **tags**, because those are the three faces the DEPOT tray runs along. The other four (both PEG FACE tags and the peg-face-side SOCKET FACE tag on each side) have no tray in front of them at any height. **beneath the tag panels on all four faces.**
- §1.3, consequence 1: ~~Final approach is unaffected for a camera near the top of the recommended band. At 20 in, the whole target is visible within 15 in, which covers close-range alignment only. At the DEPOT lip, the final approach can lose the bottom edge of one tag. The BASE DEPOT runs along every CRAG face, so BUMPERS stop at its lip, 16.75 in off the face, and a camera on the ROBOT is never nearer to a CRAG tag than that. Seeing the whole target past a CELL standing on end against the face beneath the tag needs a camera about 20.5 in high at 16.75 in of range (a camera over the BUMPER face), or 21.8 in at 19.75 in (a camera at the FRAME PERIMETER), just above the recommended band. Below that height, whether the target is clipped depends on where the CELL stands: at 16.75 in of range, a camera at 18 in is clear once the CELL stands 0.9 in off the face, and a camera at 16 in only once it stands 3.0 in off. The as-built tray floor stands every SUPPLY 0.25 in higher than the carpet, which costs about a third of every range in this table. Below about 16 in, a single camera sees the bottom edge clipped from almost any useful range whenever a CELL is stood on end in front of that face, so a design committed to one low camera should place it at 18 in or higher and treat anything beyond about 10 in of range as localization only. A design committed to one low camera should therefore place it at 18 in or higher and rely on the face's tag pair (item 3) rather than on either tag alone.~~
- §1.3, consequence 4: Add a 5 × 14 in occluder standing ~~at~~ **in** the DEPOT **against the face** in front of one ~~SHELF FACE tag (§6.8)~~ **CRAG tag, on any face (§6.8)**, and confirm that the pose solution survives on the remaining tags.
- §5.1, item 1: ... the DEPOT lip (4 in), **SCREE ridges (1.75 in)**, or a crowned CACHE CRATE standing in the DEPOT ...
- §5.2, primary camera: If it is the ROBOT's only camera, put it at **18 in or higher** within that band; ~~below about 16 in it cannot see over band; the lower it is, the more positions of~~ an O2 CELL stood on end in the DEPOT **hide the bottom edge of a CRAG tag from the lip (§1.3).** New bullet: **SCREE: a ROBOT crossing a SCREE PATCH pitches and rolls on its 1.75-in ridges, and a pose estimator that assumes the ROBOT sits flat on the carpet turns that tilt into position error. Discard or down-weight vision measurements taken while the gyro shows the ROBOT tilted.**
- §6.6, new paragraph before the SHELF FACE table: Every CRAG face has the BASE DEPOT in front of it, so no goal pose can bring the ROBOT's BUMPERS nearer to a face plane than the DEPOT lip's outer face, 16.75 in out. Every feature in the tables below is reached across the tray: with the FRAME PERIMETER 19.75 in off the face, a Shelf slot center is 12.75 in of horizontal extension away, a socket rim 11.75 in, a Low or Mid Peg tip 12.68 in and a High Peg tip 26.68 in (FIELD-CAD-PACKAGE §3). BUMPERS at the lip of the ALLIANCE's own CRAG are inside its CRAG APRON, where Game Manual rule G404 allows up to 30 in of extension rather than 18 in.
- §6.8, occlusion case: place a 5 × 14 in cylinder upright on the DEPOT tray floor (Z = 0.25) **0.25)**, **against the face** immediately in front of one ~~SHELF FACE tag;~~ **CRAG tag, and repeat it on a SOCKET FACE and on the PEG**

FACE: the tray runs beneath the tags on all four faces.

- **\$7, quick reference, new row: BASE DEPOT lip | on every CRAG face, 16.75 in out from the face plane: the nearest a ROBOT's BUMPERS come**

The tag poses, `apriltag-field-layout.json` and the ± 0.15 -in CRAG tag height tolerance are unchanged.

Rationale: the DEPOT is now in front of every CRAG tag, which moves the camera-height guidance for the upright-CELL case from one face to four.

B12. Onshape field generator (`03-field/onshape/SummitPushField.fs`) and its README

The generator builds each BASE DEPOT as a closed ring around all four faces (floor, lip with R0.25 top edges, and an entry chamfer mitred at the corners) and adds a **SCREE** option (on by default) that builds the 18 SCREE PATCHES: four half-round HDPE ridges each (R1.75), at ± 4.0 and ± 12.0 in from the patch center, cut square at the 30×30 in patch boundary, in the ridge direction of the patch table. The FIELD centerline tape is broken at each CRAG between the inner edges of its APRON tape, and only the CENTER CACHE band's side lines are laid. A new material, **hdpe** (HDPE, matte, 955 kg/m^3), is used for the ridges. The feature's built-in checks cover the ring and SCREE. The modeled field now weighs about ~~3,960~~ **4,330 lb**, of which the 63 SUPPLIES are 94.5 lb **and the 18 SCREE PATCHES about 320 lb**.

A field built with an earlier generator does not have SCREE or the DEPOT ring. Paste the new `SummitPushField.fs` into the Feature Studio and regenerate.

Rationale: the generator carries A6, A11 and A14 into the Onshape model.

B13. Kit README (`participants/README.md`), modeling notes

~~Model the Blue half once, then pattern it 180° about field center for Red. Do not model the Red CRAG or HEADWALL separately.~~

- **Model the BASE DEPOT as the closed ring it is: four legs and four corner squares around the whole CRAG. Its lip sets how far every placement on the CRAG must reach.**
- **Model SCREE as part of the FIELD: each SCREE PATCH is a 30×30 in square of four straight half-round ridges (radius 1.75 in, so 1.75 in tall, on an 8.0-in pitch) sitting directly on the carpet. Take the nine Blue patch centers and ridge directions (+45, -45 or Y) from the SCREE PATCH table in `03-field/FIELD-CAD-PACKAGE.md` §1.4, and cut each ridge square at its patch boundary.**
- **Model the Blue half once, then pattern it 180° about field center for Red. Do not model the Red CRAG, HEADWALL or SCREE PATCHES separately.**

C. CADathon brief (`02-cadathon/CADATHON-BRIEF.md`)

C1. §4.1, extension statement and legality checks

Also state the FRAME PERIMETER (≤ 120 in), starting configuration height (≤ 42 in), and maximum horizontal extension (~~≤ 18 in beyond the FRAME PERIMETER~~); **beyond the FRAME PERIMETER (≤ 18 in, or ≤ 30 in over the ROBOT's own ALLIANCE's CRAG APRON while any part of its BUMPERS intersects that APRON; G404 and R105), each with the CAD measurement that proves it.**

Legality checks: ... every MECHANISM at its commanded maximum reach within the ~~18-in limit~~; **30-in limit, and a stated means of holding the ROBOT to 18 in whenever no part of its BUMPERS intersects its own CRAG APRON, and of keeping any reach beyond 18 in over that APRON (a software limit tied to the ROBOT's position, or a mechanism that can exceed 18 in only toward the CRAG); ...**

Rationale: as A21 and A24.

C2. §4.2, item 2 (Robot architecture rationale)

New sentences after "The high tier ... requires continuous elevators or multi-stage arms.": **Reach matters as much as height. The BASE DEPOT runs under every face of the CRAG, so BUMPERS stop at its lip and the FRAME PERIMETER stands 19.75 in off the face. Both sockets on each SOCKET FACE, Low and Mid, are then 11.75 in of extension away. The High Peg roots, on the spire 14.0 in inboard of the PEG FACE, are 33.75 in away, beyond any legal extension, so a ROPE COIL is released over a High Peg tip, 26.68 in of extension away, and drops onto the peg (a release from rest is not LAUNCHING under G502). That reach is legal only under the 30-in limit that applies while any part of the ROBOT's BUMPERS intersects its own CRAG APRON (G404), and the design must hold the 18-in limit everywhere else.**

... and include the drivetrain selection ~~rationale~~: **rationale, including whether the ROBOT crosses SCREE (which needs at least 2.0 in of clearance under the whole frame and wheels of about 6 in or larger) or weaves between the SCREE PATCHES.**

Rationale: the binder should address the new reach and the drivetrain choice SCREE forces.

C3. Appendix A, the BASECAMP BOT

- **A.1, Drivetrain (new sentence):** It has too little clearance under its frame to cross SCREE, so it weaves between the SCREE PATCHES through the 41–58-in gaps instead of crossing them.
- **A.1, Scoring without a lift (new sentence):** The tray surrounds the CRAG, so the robot can tip a CACHE CRATE in on any of the four faces.
- **A.2 (new sentence before the last):** Its DEPOT points come from about eight CACHE CRATES, less than half of the tray's single-layer capacity of about 18 (Manual §3.3.4), so its partners still have room in the DEPOT.
- **A.3, rung 1:** the BASE DEPOT's ~~corner arm~~ **DEPOT** runs under the ~~Low Socket~~, **both sockets on every SOCKET FACE**, so BUMPERS stop **at its lip**, 16.75 in off the ~~SOCKET FACE~~ **face**, and the ~~rim~~ **each rim, Low or Mid**, is 11.75 in of extension away, ~~compared with the 5.0 in the Mid Socket allows away~~ (FIELD-CAD-PACKAGE §2.3).
- **A.3, rung 2 (new sentence):** The Mid Pegs overhang the PEG FACE leg of the DEPOT, and their tips are 12.68 in of extension away.
- **A.3, rung 3:** Nothing above the LEDGE RUNG is reachable from the ~~carpet~~: **carpet: the 30-in allowance of G404 applies only while a ROBOT's BUMPERS intersect its own CRAG APRON, far from the HEADWALL, so the 18-in limit holds there.**

Rationale: the reference design and its ladder follow the ring, SCREE and the scoped extension limit. The BASECAMP BOT's scoring is unchanged.

C4. Header note (editorial)

Amended by Team Update 01 (~~05-team-updates/TEAM-UPDATE-01.md~~): **Updates 01 and 02** (~~05-team-updates/TEAM-UPDATE-01.md~~ **and** ~~05-team-updates/TEAM-UPDATE-02.md~~).